

EE 102 Week 3, Lecture 1 (Fall 2026)

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1 Goals for today's lecture

- Periodicity of signals.
- The signals: the complex exponential, the unit step, and impulse.
- The unit step function as a switch and an accumulator.
- The unit impulse function as an exciter and a sampler.
- The complex exponential signal as a sinusoid and a phasor.
- Applications of the timeless trio in real-world signal processing.

2 Reminder: Even and odd signals

Recall that a mathematical function is called even if $f(-t) = f(t)$ for all t and odd if $f(-t) = -f(t)$ for all t . Examples of even functions include $\cos(t)$, t^2 , and $|t|$. Examples of odd functions include $\sin(t)$, t^3 , and the sign function $\text{sgn}(t)$.

Definition 1. A signal $x(t)$ is even if $x(-t) = x(t)$ for all t and odd if $x(-t) = -x(t)$ for all t .

Proposition 1. Any signal $x(t)$ can be uniquely decomposed as the sum of an even signal $x_e(t)$ and an odd signal $x_o(t)$.

Proof. Let $x_e(t) = \frac{x(t)+x(-t)}{2}$ and $x_o(t) = \frac{x(t)-x(-t)}{2}$. Then, we have

$$x_e(-t) = \frac{x(-t) + x(t)}{2} = x_e(t)$$

so $x_e(t)$ is even. Similarly,

$$x_o(-t) = \frac{x(-t) - x(t)}{2} = -x_o(t).$$

which is odd. Now, we can see that

$$x_e(t) + x_o(t) = \frac{x(t) + x(-t)}{2} + \frac{x(t) - x(-t)}{2} = x(t).$$

To prove uniqueness, we again use the proof by contradiction method. Assume, to the contrary, that there exist another pair of even and odd signals $x'_e(t)$ and $x'_o(t)$ such that $x(t) = x'_e(t) + x'_o(t)$. Then, we have

$$x_e(t) - x'_e(t) = x'_o(t) - x_o(t).$$

The left side is even (difference of two even functions), and the right side is odd (difference of two odd functions). The only function that is both even and odd is the zero function. Therefore, we have $x_e(t) - x'_e(t) = 0$ and $x'_o(t) - x_o(t) = 0$, which implies that $x_e(t) = x'_e(t)$ and $x_o(t) = x'_o(t)$. Hence, the decomposition is unique. $\square$

3 Periodic signals

Definition 2. A signal $x(t)$ is periodic if $\exists T_0 > 0$ such that $x(t + T_0) = x(t)$ for all t . The smallest such T_0 is the *fundamental period* of the signal.

In more verbose language, we say that a signal is a periodic signal if we can find a time shift T_0 such that shifting the signal by T_0 does not change the signal. If such a T_0 does not exist, then the signal is aperiodic. For example, $x(t) = \sin(t)$ is periodic. To find the fundamental period, we would have to find the smallest shift T_0 to satisfy $\sin(t + T_0) = \sin(t)$ for all t . Using the periodicity of the sine function, we know that $\sin(t + 2\pi) = \sin(t)$ for all t . Thus, $T_0 = 2\pi$ is the fundamental period of $\sin(t)$. Note that $T_0 = 4\pi$ also satisfies the periodicity condition, but it is not the fundamental period since it is not the smallest such T_0 . In fact, any integer multiple of 2π would satisfy the periodicity condition, but we are only interested in the smallest such value for the fundamental period.

3.1 Why periodic signals?

Just like many other concepts in signal processing, periodic signals are also a mathematical convenience! Intuitively, it is clear that if something repeats over and over again, then we can analyze just one cycle of it and extend the results to the entire signal. Therefore, studying periodic signals is often a good starting point. But you may wonder — what if the signal is not periodic? Read on.

3.2 Periodic extensions

When a signal is defined on a finite interval (e.g., a single cycle), it is often useful to *periodically extend* it by repeating that interval end-to-end. This makes the time-averaged power

well defined and makes symmetries/harmonics easier to see.

3.3 Example: Periodic or not?

Consider the following three signals. Our goal is to find out whether they are periodic or not. If they are periodic, we will report the fundamental period of these signals.

1. $x(t) = \cos(t)$
2. $x(t) = \cos(t)$ for $t \geq 0$ and $x(t) = -\sin(t)$ for $t < 0$
3. $x(t) = e^{j\omega t}$ where $\omega \neq 0$

How to prove periodicity? We can simply *propose a T_0 and check* if the periodicity condition is satisfied. If you cannot find a T_0 , that does not mean that the signal is aperiodic — it just means that you have not found the right T_0 yet! To prove aperiodicity, you have to show that no such T_0 exists. This is often done by contradiction. Proofs by contradiction is an important mathematical trick where you assume that the statement you want to prove is false and then show that this assumption leads to a contradiction (something that will be *obviously* incorrect). This implies that the original statement must be true. You will find the signal transformation properties useful in proving (a)periodicity. Finally, if you can exploit the properties of the given signal, then you will find a much easier path to the proof.

For the first example, we know that the cosine function is “oscillatory”, which indicates that it is probably periodic. Let’s try to find a T_0 such that $\cos(t + T_0) = \cos(t)$ for all t . You might propose a $T_0 = \pi$ and observe that $\cos(t + \pi) = -\cos(t)$, which is not what we were looking for. So, $T_0 = \pi$ is not a good choice for T_0 . Let’s try one more time. Propose $T_0 = 2\pi$. Then, compute $\cos(t + 2\pi) = \cos(t)$, which is a valid choice. To check if it is the fundamental period, we can see that any integer multiple of 2π would also satisfy the periodicity condition, but 2π is the smallest such value. Therefore, the fundamental period of $x(t) = \cos(t)$ is $T_0 = 2\pi$. Note that you can use trigonometric identities to help you prove periodicity in an alternate way too.

For the second signal, sketch the signal to first get an intuition about whether it is periodic or not. You will see that the signal is a cosine wave for $t \geq 0$ and a negative sine wave for $t < 0$. The two parts do not match at $t = 0$, which indicates that the signal is not periodic. To prove this, we can use contradiction. Assume that the signal is periodic with period T_0 . Then, we have $\cos(t + T_0) = x(t + T_0) = x(t)$ for all t . Now, consider the case when $t = -\frac{T_0}{2}$. Then, we have $\cos(-\frac{T_0}{2} + T_0) = \cos(\frac{T_0}{2}) = x(-\frac{T_0}{2}) = -\sin(-\frac{T_0}{2}) = \sin(\frac{T_0}{2})$. This is clearly not true! So, our assumption that the signal is periodic must be false. Therefore, the signal is aperiodic.

For the third signal, we can use the properties of the complex exponential function to prove periodicity. Write $x(t + T_0) = e^{j\omega(t+T_0)} = e^{j\omega t} e^{j\omega T_0}$. If we can find a T_0 such that $e^{j\omega T_0} = 1$, then we have periodicity. This is satisfied if $T_0 = \frac{2\pi}{\omega}$. Therefore, the fundamental period of $x(t) = e^{j\omega t}$ is $T_0 = \frac{2\pi}{\omega}$.

3.4 Power and energy of periodic signals

We can revisit the power and energy definitions for periodic signals. If x is periodic with period T_0 , then the time-average power is well defined and can be computed over any interval of length T_0 . Note that for a finite-duration input, E_∞ is finite and $P_\infty = 0$ (time average over an unbounded window goes to zero) whereas E_∞ is finite because we have finite-duration signal. For periodic signals, we have

$$P_\infty(x) = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} |x(t)|^2 dt \quad (\text{independent of } t_0), \quad E_\infty(x) = \infty \text{ unless } x \equiv 0.$$

Thus, periodic signals are *power signals* (finite power, infinite energy).

Energy and power for a periodic input

If $x(t)$ is periodic with fundamental period T_0 , then $y_d(t)$ is also periodic with the *same* T_0 (memoryless mapping preserves period). Hence

$$E_\infty(y_d) = \int_{-\infty}^{\infty} |y_d(t)|^2 dt = \infty, \quad P_\infty(y_d) = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} |y_d(t)|^2 dt \text{ (finite).}$$

4 The unit step function

When we defined signals informally, we discussed how any mathematical function from your calculus class could be a signal as long as it represents something physical. Then it will not come as a surprise that for physical system applications, we would usually be interested only in positive values of time and we would want our signal to take the value of zero for all $t < 0$. Since we are extending the general concept of mathematical functions (which are defined for all t), it is important to have a mathematical way to write signals that are zero for $t < 0$. The unit step function does exactly that! Formally, we define the unit step function as

Definition 3. The unit step is a function defined as

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}.$$

4.1 The unit step as a switch

As you can see in Figure 1, it is a discontinuous function that “steps” from 0 to 1 at $t = 0$. You may find the unit step function with different names: like the Heaviside function, or the ultrasensitive switch function.

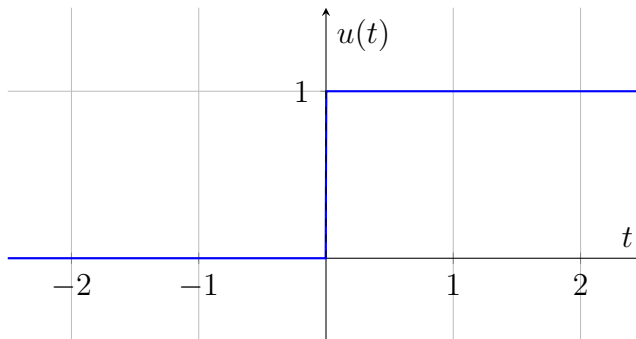


Figure 1: The unit step function $u(t)$.

So, any function that is zero for $t < 0$ can be written as the product of the unit step function and another function that defines the behavior of the signal for $t \geq 0$. For example, if we have a signal that is zero for $t < 0$ and equals $f(t)$ for $t \geq 0$, we can write it as $x(t) = f(t)u(t)$.

4.2 Example: A sinusoidal audio wave that starts at zero time

If $f(t) = A \sin(\omega t + \phi)$, then we can write the signal as

$$x(t) = A \sin(\omega t + \phi)u(t).$$

This signal is zero for $t < 0$ and equals a sinusoidal wave for $t \geq 0$. The unit step function effectively “switches on” the sinusoidal wave at $t = 0$. But did we *really* need the step function here? You can argue that we could have defined all signals using two cases,

$$x(t) = \begin{cases} f(t), & t \geq 0 \\ 0, & t < 0 \end{cases}$$

but this type of definition will quickly get cumbersome. So, yet again, we are introducing a mathematical object to make our lives easier, at least in the long run (at the moment, it may seem that we are making our life harder by learning another new function). For the specific sinusoidal example, the alternative way to define it is using a piecewise function that would need two different cases:

$$x(t) = \begin{cases} 0, & t < 0 \\ A \sin(\omega t + \phi), & t \geq 0 \end{cases}.$$

We would prefer $x(t) = A \sin(\omega t + \phi)u(t)$ over the piecewise definition so that we can work with just a single expression.

4.3 Unit step as a general switch

Beyond the switching behavior of unit step, we can also use it to define arbitrary “pulse” signals and also other piecewise continuous signals. For example, we can define a rectangular pulse of width τ as

$$p_\tau(t) = u(t) - u(t - \tau).$$

This pulse is 1 for $0 \leq t < \tau$ and zero otherwise (can you prove this without relying on sketching?). It is also possible to write the equation of the pulse signal by using a time reversal:

$$p_\tau(t) = u(t) + u(\tau - t) - 1$$

The sketch in both cases is the same: a pulse that starts at $t = 0$ and ends at $t = \tau$. More generally, we can write a pulse that starts at t_1 and ends at t_2 as

$$p_{t_1, t_2}(t) = u(t - t_1) - u(t - t_2).$$

See Figure 2 for a sketch of the rectangular pulse that starts at t_1 and ends at t_2 .

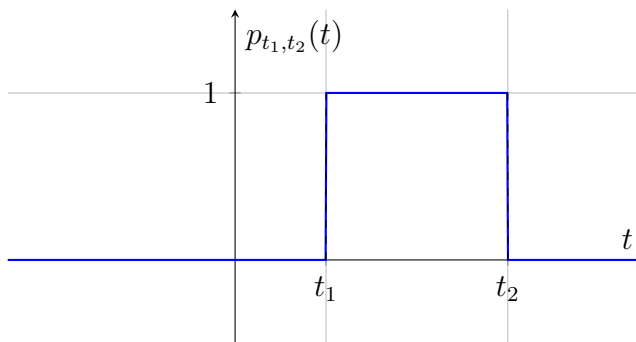


Figure 2: A rectangular pulse $p_{t_1, t_2}(t)$ that is 1 for $t \in [t_1, t_2)$ and 0 otherwise.

The pulse function provides us with more control over when the signal “turns on” and “turns off” — in Figure 2, it turns on at $t = t_1$ and turns off at $t = t_2$. We would expect such knobs of any modern switching system!

We can also define a triangular pulse of width 2τ that is zero for all $t < 0$ and $t > 2\tau$, and peaks at $t = \tau$. This triangular pulse would increase linearly from 0 to 1 in the interval $[0, \tau]$ and then decrease linearly from 1 to 0 in the interval $[\tau, 2\tau]$. By writing the equations of the two linear segments and then combining with unit step to decide when would each

linear segment be “active”, we can write the equation of the triangular pulse. Let’s build it step-by-step. We write the two parts that “activate” the two linear segments using unit step functions:

$$w_1(t) = \underbrace{u(t) - u(t - \tau)}_{\text{active on } [0, \tau)}, \quad w_2(t) = \underbrace{u(t - \tau) - u(t - 2\tau)}_{\text{active on } [\tau, 2\tau)}.$$

now, we define the two linear segments:

$$r(t) = \underbrace{\frac{t}{\tau}}_{\text{linear rise}}, \quad f(t) = \underbrace{2 - \frac{t}{\tau}}_{\text{linear fall}}.$$

Then, the triangular pulse is obtained by gating each linear segment with its corresponding window and then adding the two gated segments:

$$tr_\tau(t) = \underbrace{r(t)}_{\text{rise}} \underbrace{w_1(t)}_{\text{gate } [0, \tau)} + \underbrace{f(t)}_{\text{fall}} \underbrace{w_2(t)}_{\text{gate } [\tau, 2\tau)}.$$

$$tr_\tau(t) = \frac{t}{\tau} [u(t) - u(t - \tau)] + \left(2 - \frac{t}{\tau}\right) [u(t - \tau) - u(t - 2\tau)].$$

See Figure 3 for a sketch of the triangular pulse.

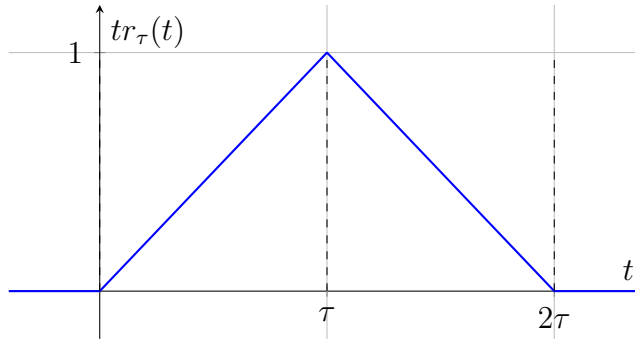


Figure 3: Triangular pulse of width 2τ : zero outside $[0, 2\tau]$, peak 1 at $t = \tau$.

The triangular pulse provides a smoother transition between the off and the on states compared to the rectangular pulse and is constructed using the unit step function as well. So, the unit step function is not just a simple switch it is also a building block for constructing more complex signals. The next “pulse” type signal that you should try is a trapezoidal pulse!

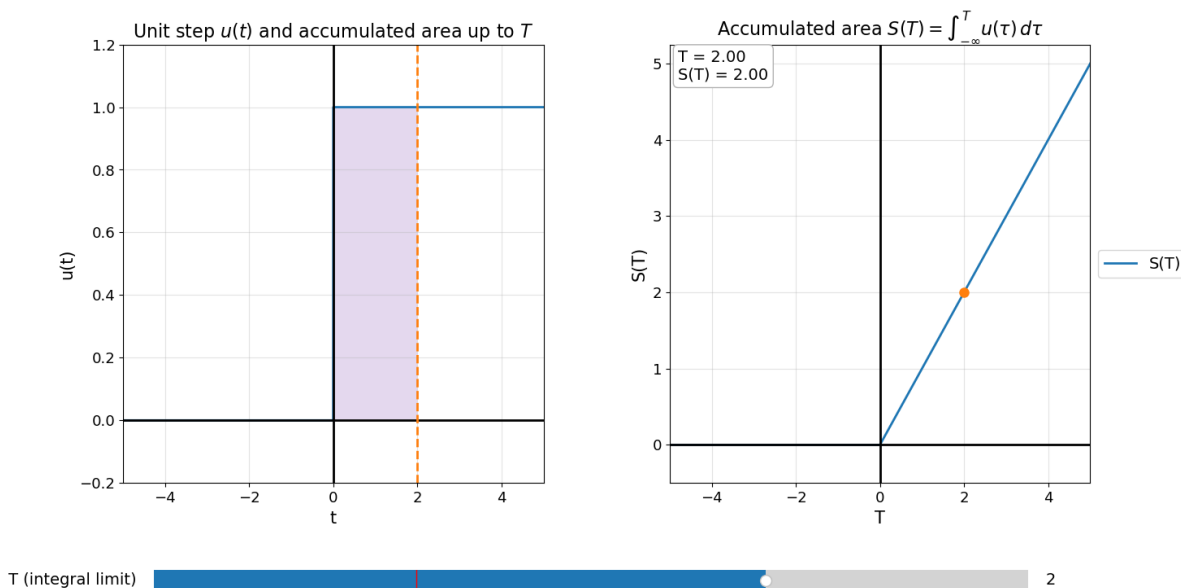


Figure 4: The unit step function $u(t)$ (left) and its accumulated area, the ramp function $r(t)$ (right). The slider below sets the upper limit of integration T .

4.4 The unit step as an accumulator

The unit step function can also be viewed as an accumulator. If we integrate the unit step function, we obtain the ramp function (Problem 2.1 in Problem Set #2):

$$r(t) = \int_{-\infty}^t u(\tau) d\tau = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}.$$

This ramp function increases linearly for $t \geq 0$ and is zero for $t < 0$. It effectively accumulates the area under the unit step function. See Figure 4 for a sketch of the unit step function and its accumulated area (the ramp function) on the right side. A virtual manipulative is available for you to explore this concept interactively on the course Github page¹.

5 The unit impulse function

The unit impulse function is our second of the “timeless trio” of signals — one of the three most important signals in signal processing. In other fields, such as physics, it is also known

¹Here is the [link](#) for virtual manipulative on Github for unit step as an accumulator

as the Dirac delta function. In discrete mathematics, it is known as the Kronecker delta function. So, the same mysterious function has many different names! The reason is clear — it is a very useful mathematical object, while not even being a function in the traditional sense! We will define it formally later, but for now, let's understand it informally. You only need to remember two properties about the impulse function:

- It is zero at all times except at $t = 0$, that is $\delta(t) = 0$ for all $t \neq 0$.
- It has an area of 1 under its curve, that is, $\int_{-\infty}^{\infty} \delta(t) dt = 1$.

How is that possible? If a signal is zero everywhere except at one point, then there must be something unique happening at that point. To build intuition for this function, consider the rectangular pulse we defined earlier, with a slight modification. Let's define a rectangular pulse of width ϵ and height $\frac{1}{\epsilon}$ around the origin:

$$p_{\epsilon}(t) = \frac{1}{\epsilon} \left[u\left(t + \frac{\epsilon}{2}\right) - u\left(t - \frac{\epsilon}{2}\right) \right].$$

A sketch of this pulse is shown in Figure 5.

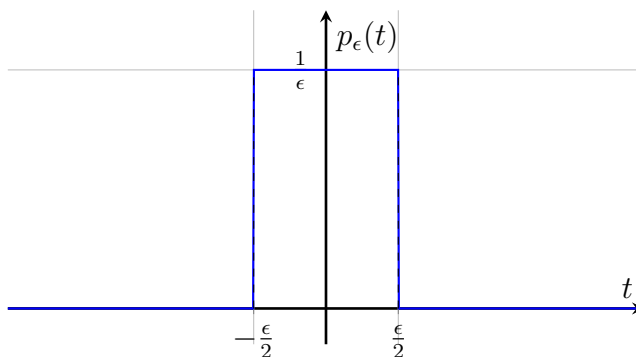


Figure 5: Rectangular pulse $p_{\epsilon}(t) = \frac{1}{\epsilon} \left[u\left(t + \frac{\epsilon}{2}\right) - u\left(t - \frac{\epsilon}{2}\right) \right]$ of width ϵ centered at the origin.

As you can see in the figure, the area of the rectangle is equal to 1 for all values of ϵ . But this does not satisfy the two properties of the impulse function listed above as there are points $t \neq 0$ where the pulse is non-zero. So, we need to modify this pulse further. As we make ϵ smaller, visualize how the rectangle becomes taller and narrower, while still maintaining an area of 1. You can interactively explore this concept using the virtual manipulative available on the course Github page².

²Here is the [link](#) for virtual manipulative on Github for impulse approximation using a pulse. Additionally, you can approximate an impulse using a Gaussian function, see [here](#).

5.1 Impulse as the limit of a rectangular pulse

Formally, we can write the unit impulse function as the limit of the rectangular pulse as ϵ approaches zero:

$$\delta(t) = \lim_{\epsilon \rightarrow 0} p_{\epsilon}(t) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left[u\left(t + \frac{\epsilon}{2}\right) - u\left(t - \frac{\epsilon}{2}\right) \right].$$

Despite the definition above, it is not possible to write a closed-form expression for the impulse function because it is not defined at $t = 0$ and is zero everywhere else. The only quantifiable property that we know so far is that the area under the curve of a delta function is equal to 1. To visually describe an impulse function, we draw an arrow pointing upwards at the point at which the impulse is located. That is, if we have $\delta(t)$, we draw an arrow at $t = 0$; if we have $\delta(t - t_0)$, we draw an arrow at $t = t_0$. The height of the arrow is not important, but we label it with a number to indicate the area under the impulse. For example, if we have $A\delta(t - t_0)$, we draw an arrow at $t = t_0$ and label it with A to indicate that the area under the impulse is equal to A .

In practice, we can never generate a true impulse function, we can only get infinitesimally close to it as we keep making the width of the rectangular pulse infinitesimally close to zero and the height of the pulse close to infinity. Even though this may sound needlessly confusing, it provides us with a very powerful mathematical tool. One example is discussed next.

5.2 Impulse as a time-sampler

The impulse function can sample any test function at a specific point in time. Note that if we write $f(t)\delta(t)$, the product will be zero for all $t \neq 0$ because $\delta(t)$ is zero for all $t \neq 0$. The only point where the product is non-zero is at $t = 0$. So we can write $f(t)\delta(t) = f(0)\delta(t)$. This is also an impulse function located at $t = 0$ with an area of $f(0)$. Now if we integrate this product over all time, we get

$$\int_{-\infty}^{\infty} f(t)\delta(t)dt = f(0) \int_{-\infty}^{\infty} \delta(t)dt = f(0) \cdot 1 = f(0).$$

This property is known as the sifting property of the impulse function. By integrating any test function multiplied by an impulse function, we can extract the value of the test function at the location of the impulse $\rightarrow$ we have sampled that function! More generally, if we have an impulse located at $t = t_0$, we can write this impulse as $\delta(t - t_0)$. Then, by the same reasoning as above, we can show that

$$f(t_0) = \int_{-\infty}^{\infty} f(t)\delta(t - t_0) dt.$$

To prove the above, write the integral as

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_0) dt = f(t_0) \int_{-\infty}^{\infty} \delta(t - t_0) dt = f(t_0) \cdot 1 = f(t_0).$$

We obtained the latter equality by observing that $\delta(t - t_0)$ is zero at every point other than $t = t_0$. Since $f(t_0)$ is independent of t , we can take it outside the integral. The remaining integral is equal to 1 because the area under the impulse function is equal to 1. This property makes the impulse function a powerful tool in signal processing and system analysis.

In formal mathematical analysis, the above is not seen as a property but is instead used to *define* the impulse function.

Definition 4. The unit impulse function $\delta(t)$ is defined as the function for which the area under curve of its product with any test function $f(t)$ that is continuous at $t = 0$, is equal to the value of the test function at the time instant at which the impulse is located ($t = 0$ for $\delta(t)$). That is,

$$\int_{-\infty}^{\infty} f(t)\delta(t)dt = f(0). \quad (1)$$

5.3 Relationship between unit step and unit impulse

If you look back at the unit step function, you will notice that it is discontinuous at $t = 0$. So, can we define the differentiation of unit step function with time, du/dt ? Generally, the answer will be no since the derivative of a function is not defined at points where the function is discontinuous. Let's try an alternate approach. Consider the following integral:

$$\int_{-\infty}^{\infty} f(t)\frac{du(t)}{dt}dt.$$

We can evaluate this integral using integration by parts to write

$$\begin{aligned} \int_{-\infty}^{\infty} f(t)\frac{du(t)}{dt}dt &= f(t)u(t)|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} u(t)\frac{df(t)}{dt}dt \\ &= f(\infty) - \int_0^{\infty} \frac{df(t)}{dt}dt \\ &= f(\infty) - [f(t)]_0^{\infty} \\ &= f(\infty) - f(\infty) + f(0). \end{aligned}$$

So, we derived that

$$\int_{-\infty}^{\infty} f(t)\frac{du(t)}{dt}dt = f(0), \quad (2)$$

which is the same as the definition of the impulse function discussed earlier — the area under the product of any test function and the impulse function is equal to the value of the test function at the location of the impulse. So, we can conclude by comparing equations (1) and (2) that

$$\frac{du(t)}{dt} = \delta(t).$$

Using a similar approach, you can also show that (Problem Set #2)

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau.$$

6 Next lecture: Complex exponential signals

We will talk about complex exponential signals in the next lecture. For now, remember the Euler's identity:

$$e^{j\theta} = \cos(\theta) + j \sin(\theta).$$

So, with a phase angle ϕ , we can write a complex exponential signal as

$$x(t) = A e^{j(\omega_0 t + \phi)} = A \cos(\omega_0 t + \phi) + j A \sin(\omega_0 t + \phi).$$

Real and imaginary parts are orthogonal sinusoids. Fundamental period $T_0 = \frac{2\pi}{\omega_0}$.

In discrete time,

$$x[n] = A e^{j(\Omega_0 n + \phi)}.$$

This is periodic iff $\frac{\Omega_0}{2\pi} = \frac{M}{N}$ with integers M, N coprime. Then the fundamental period is $N_0 = N$. Otherwise, it is *aperiodic* on $\mathbb{Z}$.